

Wide-field, High-resolution Snapshot Compressive Microscopy for Laser Materials Manufacturing

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Objectives

1. To mitigate the trade-off between optical resolution, field-of-view and imaging speed.
2. To enable the high-throughput imaging by temporal compressive imaging.

Introduction

Traditional microscopy techniques have limited functionality when it comes to capturing high-resolution images of large areas quickly. There are several ways to increase field of view (FOV) and resolution simultaneously. Direct ways involve using objective lens with high numerical aperture or increasing sensor size. Both methods require expensive hardware, leading to increased costs. Another approach is through the use of computational techniques such as Fourier Ptychography. This method involves acquiring a series of low-resolution images and computationally merging them to produce a high-resolution image. This process is time-consuming and thus fails to reconstruct high-speed scenes.

A promising approach for addressing the limited imaging speed is snapshot compressive imaging [1], which combines hardware and software to enable high-speed in a single snapshot. To mitigate the inherent trade-off between spatial resolution, FOV and imaging speed, we propose a dual-path snapshot compressive microscopy (DP-SCM). DP-SCM consists of two parallel optical paths, one optimized for high-resolution imaging and the other for wide-field imaging. By combining these dual measurements, DP-SCM can reconstruct high-resolution images over a large FOV at high imaging speed.

Snapshot Compressive Imaging

- ▶ Principle: hardware encoder and software decoder
 - ▶ The dynamic scenes are optically encoded and integrated in a snapshot, which is then fed into neural network for dynamic reconstruction.
- ▶ Advantages: high-speed, low-bandwidth and low-cost imaging

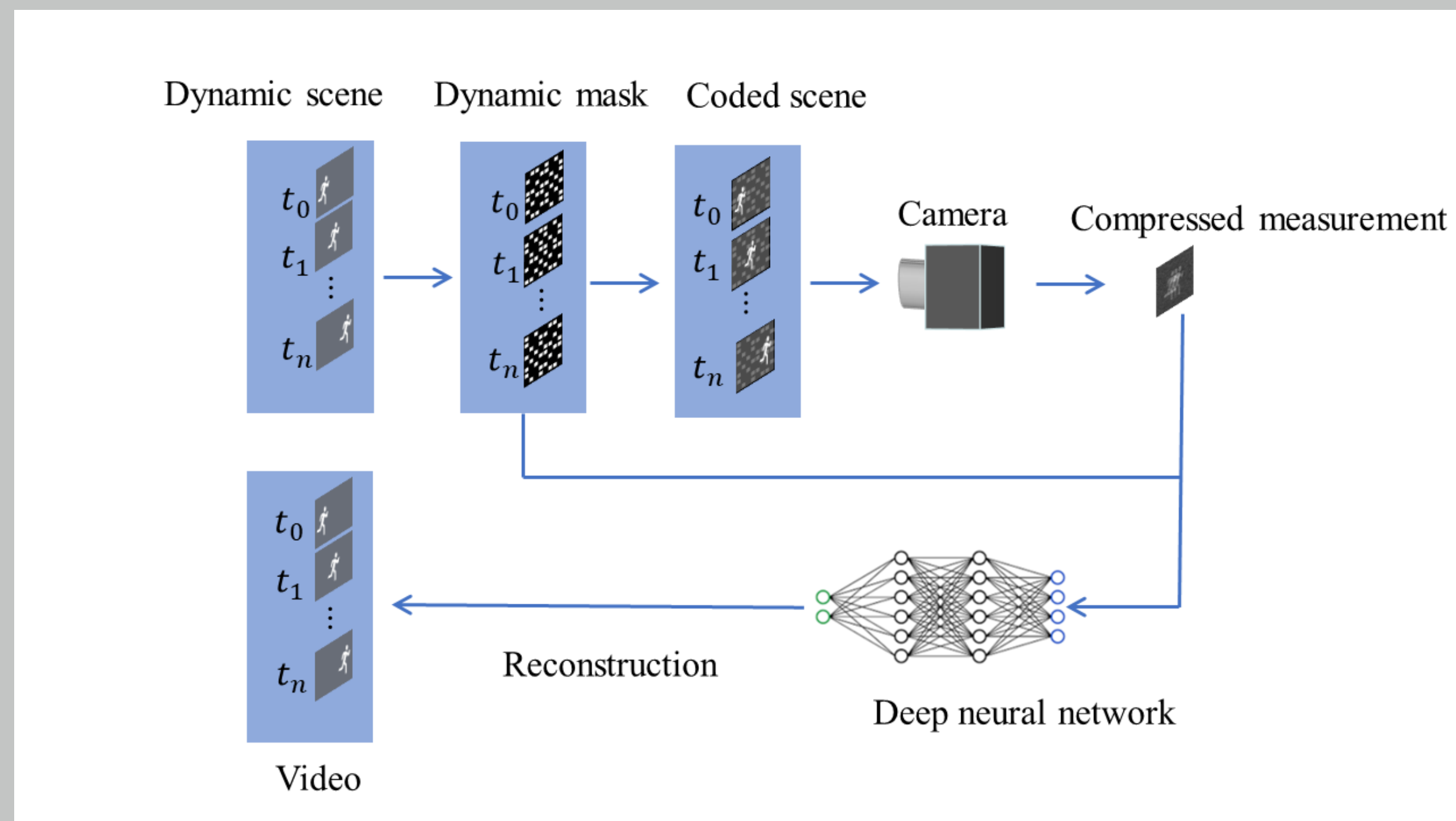


Figure 1: Principle of temporal snapshot compressive imaging

Methods

On the basis of temporal snapshot compressive imaging, we propose dual-path snapshot compressive microscopy, see Fig. 2. The DP-SCM system consists of two parallel optical paths, each optimized for a different range of resolution and FOV.

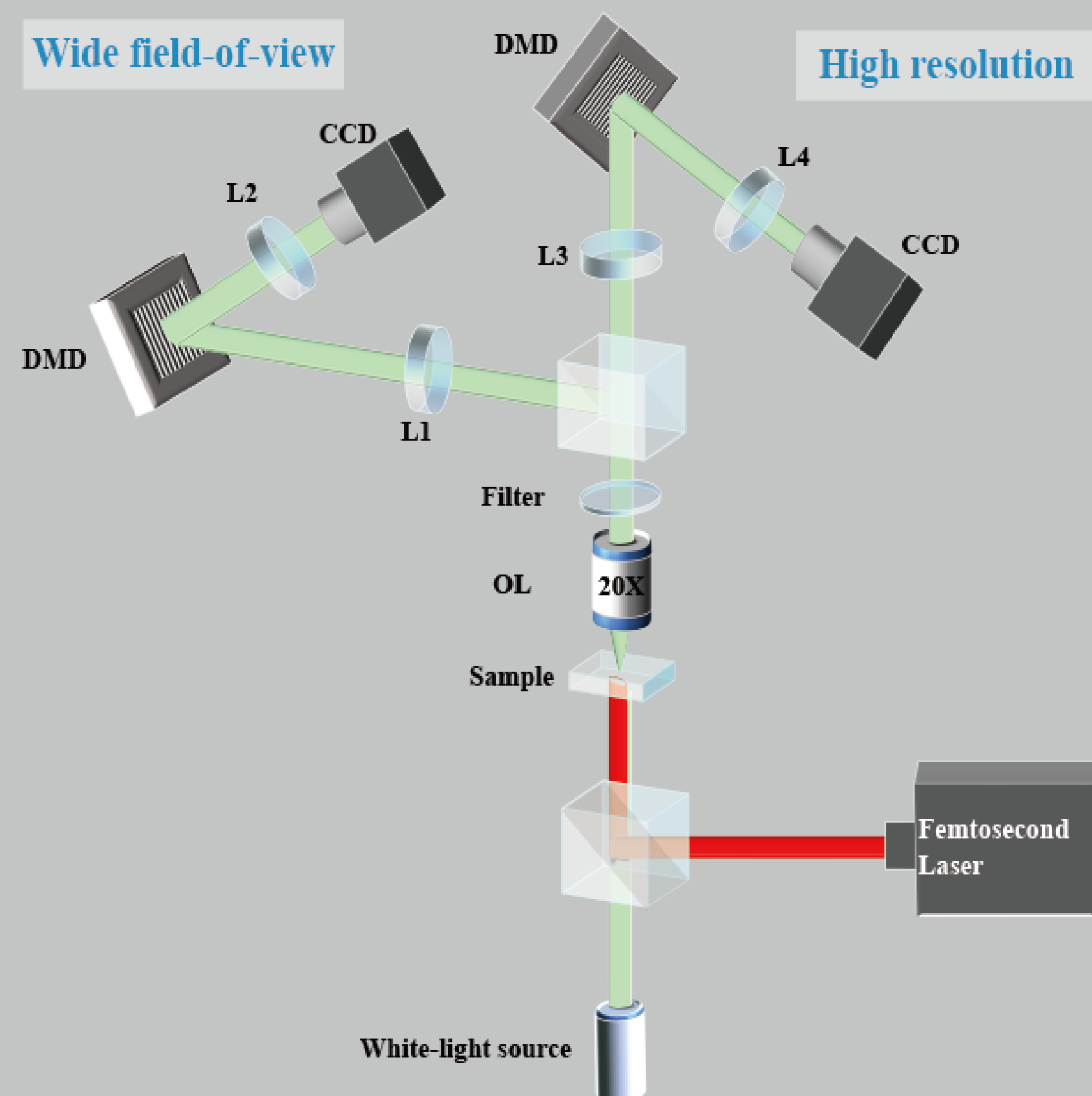


Figure 2: Dual-path wide-field, high-resolution snapshot compressive microscopy.

Mathematics

Considering that the optical encoding $\mathbf{M}(:, :, \mathbf{t})$ is binary mask, where "0" blocks light and "1" transmits light, the detected intensity can be modeled as

$$\mathbf{Y} = \sum_{t=1}^T \mathbf{U}(:, :, t) \odot \mathbf{M}(:, :, t) + \mathbf{G} \quad (1)$$

where $\mathbf{G}$ denotes the measurement noise, $\mathbf{U}$ and $\mathbf{Y}$ represent the dynamic scene and measurement respectively, now rewrite the forward model in a more concrete way,

$$y = \Phi u + g \quad (2)$$

where y and g are vectored form of $\mathbf{Y}$ and $\mathbf{G}$. Φ denotes the sensing matrix, which is a concatenation of diagonal matrices. Specially, $\Phi = [\Phi_1, \dots, \Phi_T]$, where $\Phi_t = \text{Diag}(\text{vec}(\mathbf{M}(:, :, t)))$ is a diagonal matrix with its diagonal elements composed of $\text{vec}(\mathbf{M}(:, :, t))$. We see that this is a typical inverse problem, which can be solved by means of optimization-based or deep-learning based methods.

Results

- ▶ We validated its large FOV and high resolution imaging using ultra-high resolution target. Both paths employ the same objective lens but uses a long-focal-length lens. A FOV of 2mm is attained, where even the smallest line with a linewidth of 775nm is clearly distinguishable (as shown in Fig. 2(a-b)).
- ▶ We validated snapshot compressive imaging in high-resolution path using a moving resolution target and captured the cocompressed measurement, see Fig.3(c) left. The measurement was fed into an efficient neural network for reconstruction. Building upon the remarkable achievements of convolution and Transformer [2] in computer vision, we have designed a reconstruction network that leverages the strengths of both techniques. In this way, using camera working at 50fps, setting compressed ratio (CR) to 10, dynamic imaging with high resolution and large FOV at 500fps is realized without losing reconstruction quality.

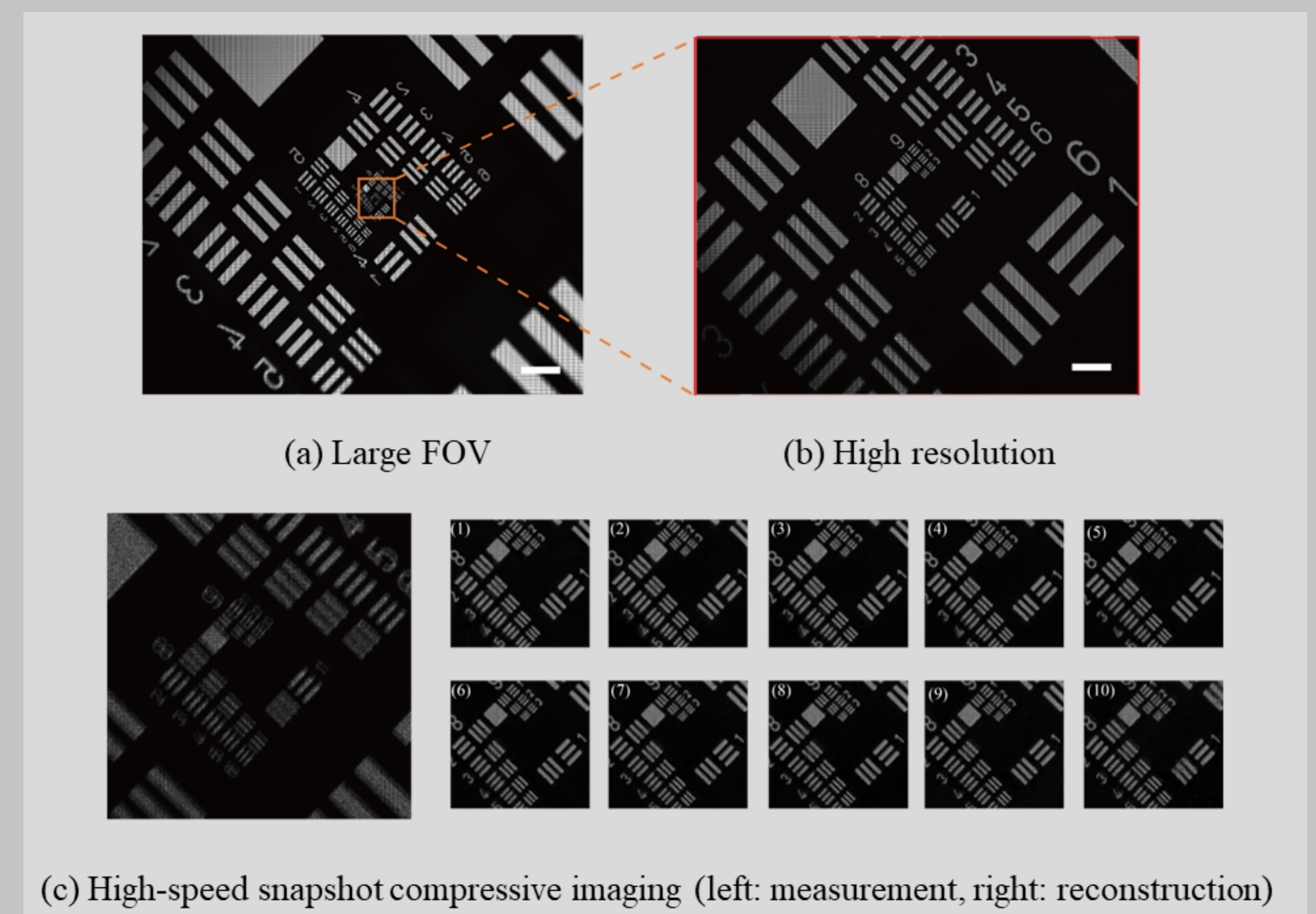


Figure 3: Wide-field, high-resolution snapshot compressive imaging. Scale bars in (b) and (c) are 250um and 25um respectively

Conclusions

Our DP-SCM approach offers a promising solution for achieving high-speed, high-resolution and wide field-of-view imaging simultaneously without sacrificing image quality, opening up new possibilities for applications in medical diagnosis, biological imaging, and materials science. We expect this method will find more applications in fields of laser materials manufacturing.

References

- [1] Xin Yuan, David J. Brady, and Aggelos K. Katsaggelos. Snapshot compressive imaging: Theory, algorithms, and applications. *IEEE Signal Processing Magazine*, 38(2):65–88, 2021.
- [2] Lishun Wang, Miao Cao, Yong Zhong, and Xin Yuan. Spatial-temporal transformer for video snapshot compressive imaging. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(7):9072–9089, 2023.

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